

CRAFT — Pitch Package

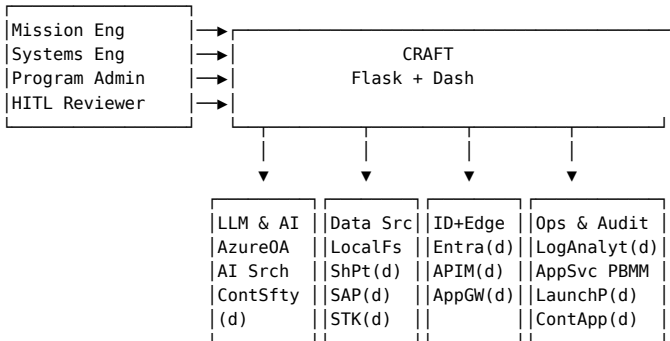
Legend

■ Actor · ◇ Gate · ☁ Cloud · [?] Terminal · [?] Store · --- Planned · <I> Protocol · Blue=Auth · Green=Audit

1. What CRAFT Is

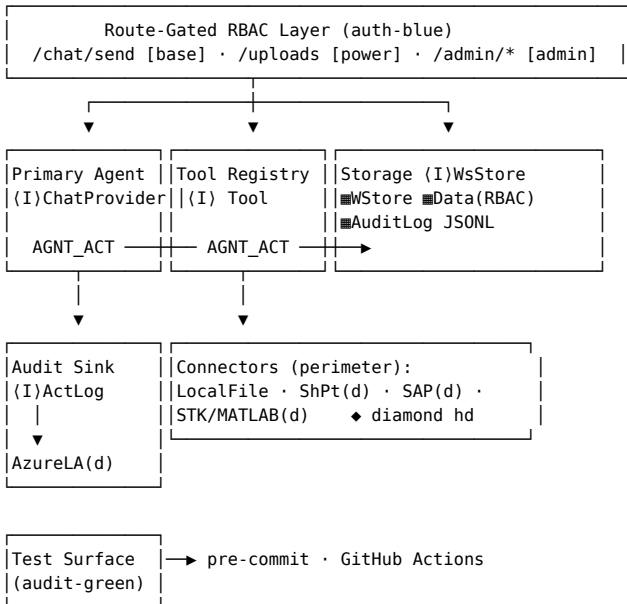
Audited retrieval-augmented AI assistant for CSA mission engineering and finance. Scratchpad MVPs the seams. Nothing ships — pattern is the point.

2a. L1 System Context



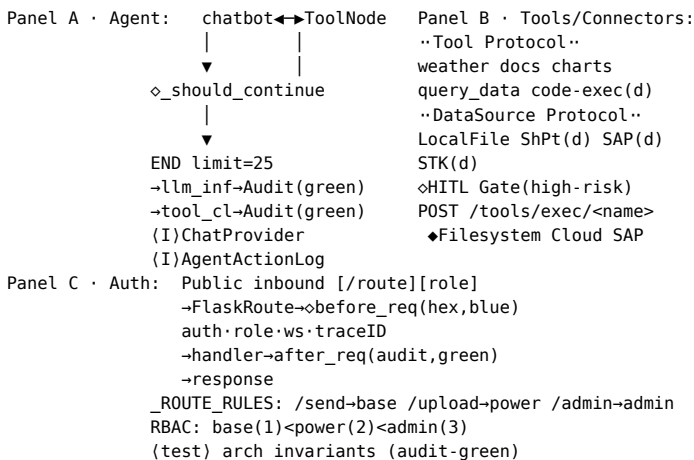
In: /chat/send [base] · /admin/* [admin]. Out: thin italic. (d)=planned/dashed.

2b. L2 Container



AGENT_ACTION (green). Connectors ♦. Test surface dotted.

2c. L3 Internals



2d. UC State Models

UC-E2 Risk ID+HITL (Eng): Req→Auth-Retrieval[002-006]→Classifier[036-040]→HITL[HITL-001]
Approve→RiskReg(d)→Audit→● Reject→●

UC-F1 Cost+HITL (Fin): Req→Auth→Retrieval(hist)[031]→CostAgg[032-034]→HITL[035]
Approve→Commit(d)→Audit→● Reject→●
UC-F4 SAP Conn (blocked): Req→Auth→SAPConn(d)[053-055]→VendorExt→Agg→Audit→●(blocker:SAP)

●=Terminal. ◇=Gate. (d)=planned. Swim lanes: User→Agent→Services→Terminal.

3. Requirements → Design

Primary agent — UMR-001/011/013/016/018/019/094/095, AAR-001/003/006, ASG-005. LangGraph ReAct · ChatProvider · recursion_limit=25. No blocker. Seam: ChatProvider. UC: E2, F1, F4.

Retrieval + RAG — UMR-002/003/004/005/006/009/010/060/067, ARR-002-007. Azure AI Search (hybrid vector+BM25) · DataSource+Tool. Blocker: IT AI Search deploy. Seam: HybridRetrievalTool. UC: E2, F1.

Audit + traceability — UMR-015/027/045/061/062/093, AAR-005, ASG-003. AgentAction envelope · AgentActionLog · JSONL→Log Analytics. Blocker: IT Log Analytics+immutability. Seam: AgentActionLog sink. UC: E2, F1, F4.

HITL approvals — UMR-021-026/035/039/044/049, HITL-001-007, ASG-002. LangGraph interrupt · Postgres queue · Dash UI. Blocker: CSA risk taxonomy. Seam: ChatProvider+interrupt. UC: E2, F1.

RBAC + data scoping — UMR-012/017/058/065/066, AAR-002/007. _ROUTE_RULES+_ROLE_LEVEL+SqliteDataStore+QueryTool.set_user. Blocker: per-tool RBAC pending. Seam: rule list appends. UC: E2, F1, F4.

Session + workspace (V3) — UMR-014/092/095. Split WorkspaceStore→SqliteStore+SessionScratchStore. Blocker: session lifecycle design. Seam: WorkspaceStore protocol. UC: dream.

Guardrails + containment — UMR-020/029/030, ASG-001/004/011. Azure Content Safety · APIM rate limit · Container Apps sandbox. Blocker: procurement. Seam: CodeExecTool. UC: dream.

Connectors + external — UMR-051-055. DataSource+MSAL · Flask REST · SAP/STK/MATLAB planned. Blocker: SAP procurement; STK/MATLAB licenses. Seam: DataSource classes. UC: F4.

Domain capabilities — UMR-007/008/031-034/036-038/040-043/046-048/050/075, ARR-001. New Tool over existing seams. Blocker: taxonomy+mission DB+Translator. Seam: uniform pattern. UC: E2, F1.

Deployment, SLA — UMR-028/056/057/059/064/069/070/074/085, ASG-006. Azure App Service PBMM · APIM+WAF · SSC LaunchPad HA · git config. Blocker: IT landing zone; LaunchPad. Seam: env-var+SQL migrations. UC: dream.

4. Use Cases

UC-E2 · **Risk ID + HITL**. Engineer uploads CADRe Part A → retrieval → LLM classifier → HITL gate → approve/reject → register write (planned).

UC-F1 · **Parametric cost + HITL**. Admin requests estimate → 5 similar missions → Euclidean distance → HITL gate before commitment.

UC-F4 · **Vendor cost + SAP**. Analyst queries vendor costs → SAP connector (DataSource planned) → extract+aggregate → audit. Blocked: SAP.

Other flows: budget scenarios, anomaly triage, bilingual reports, cross-mission compare — each = new Tool+(new DataSource). Core unchanged. No new architecture. Gated by §5.

5. Blocked Items

- **Azure AI Search (IT)** — retrieval+citations. Seam: DataSource+HybridRetrievalTool.
- **Log Analytics+immutability (IT)** — tamper-proof audit. Seam: AgentActionLog.
- **Content Safety+Container Apps (procurement)** — guardrails+sandbox. Seam: CodeExecTool.
- **Entra ID+SharePoint/SAP (IT+finance)** — CSA corpus+cost data. Seam: DataSource.
- **CSA risk taxonomy (domain)** — risk ID+HITL.
- **Historical mission DB (finance)** — cost estimation.

6. System-Health Tooling

- **ruff** — Lint (flake8+isort+pyupgrade) · pre-commit+CI
- **pyright** — Strict type check · pre-commit+CI
- **pytest+cov** — 156 tests, 91% coverage · pre-commit+CI
- **bandit** — Security scan · pre-commit
- **pip-audit** — CVE audit · pre-commit
- **detect-secrets** — No credentials in git · pre-commit

Architecture tests enforce protocol boundaries+required files per feature. Code quality: logging coverage, function length, import completeness. Suite: .pre-commit-config.yaml+.github/workflows/ci.yml. New invariant=new test. No extra infra.